ANALYSIS REPORT

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

`TE Curve Verification Pipeline` is the official offline verification surface for newly introduced TE models. A model or family is not considered verified only because its training campaign metrics improved; it must also be compared on the direction-aware `TE Curve Verification Pipeline` curve-reconstruction matrix and reviewed against visual curve evidence.

Current closeout verdict

- `TE Curve Verification Pipeline` is accepted as the canonical offline model-verification report.
- `Target A` is closed as an offline direction-qualified benchmark.
- The strongest current offline paper-derived references are `paper_retuned_best_Fw` and `paper_retuned_best_Bw`.
- The strongest current individual reference-family candidates are `rcim_retuned_GBM19_Fw` and `rcim_retuned_GBM19_Bw`.
- The strongest current repository-owned static baseline remains the `tree` / `hist_gradient_boosting` family.
- Future `TE Curve Verification Pipeline` campaigns must update this report, the source matrix, and the visual companion reports before their results are treated as accepted.

Source Package

This official report consolidates these approved `TE Curve Verification Pipeline` artifacts:

- metric matrix: `doc/reports/analysis/track2/Track 2 Directional Model Comparison.md` ;
- best-model collage report: `doc/reports/analysis/track2/best_model_collage_report/[2026-05-20]/track2_best_model_collage_report.md` ;
- best-model collage PDF: `doc/reports/analysis/track2/best_model_collage_report/[2026-05-20]/track2_best_model_collage_report.pdf` ;
- multi-model curve comparison report: `doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-20]/track2_multi_model_curve_comparison_report.md` ;
- multi-model curve comparison PDF: `doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-20]/track2_multi_model_curve_comparison_report.pdf` . Machine-readable and visual validation outputs are retained under:
- `output/validation_checks/track2_reference_comparison/` ;
- `output/validation_checks/track2_best_model_collage_report/` ;
- `output/validation_checks/track2_multi_model_curve_comparison_report/` .

Verification Rule

The repository treats direction as a first-class verification surface:

Surface	Training or Archive Scope	Evaluation Scope
`global`	forward and backward together	both directions, reported separately
`Fw` / `forward`	forward only	forward curves only
`Bw` / `backward`	backward only	backward curves only

The rule applies to paper-reference models, `RCIM Model-Bank Reproduction` faithful archives, retuned reference archives, `Wave 1` exported models, and future `TE Curve Verification Pipeline` campaign candidates.

Pipeline Coverage

Pipeline or Source	TE Curve Verification Pipeline Role	Current Status	Verification Artifact
recovered original RCIM archive	paper-original forward reference	included	directional matrix
retuned RCIM archive	current paper-derived forward and backward baseline	included	directional matrix and visual reports
`RCIM Model-Bank Reproduction` exact paper-faithful bank	source-faithful reproduction evidence	included	directional matrix and visual reports
`Wave 1` exported static baselines	repository-owned model candidates	included	directional matrix and visual reports
`periodic_mlp` explicit harmonic campaign	latest family-registry refresh	included in visual refresh	collage and overlay reports
future `TE Curve Verification Pipeline` campaigns	new verification candidates	append here when approved	matrix, collage, overlay, PDF

Current Numeric Baselines

Best Composite References

Candidate	Source	Direction	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
`paper_original_best_Fw`	`rcim_original`	forward	0.002769	0.002951	6.250	13.827
`paper_retuned_best_Fw`	`rcim_retuned`	forward	0.001839	0.002041	4.109	9.866
`track1_best_Fw`	`rcim_track1`	forward	0.003014	0.003204	6.819	11.638
`paper_retuned_best_Bw`	`rcim_retuned`	backward	0.003675	0.004284	7.572	15.645
`track1_best_Bw`	`rcim_track1`	backward	0.005027	0.005212	11.860	48.106

Strongest Individual Reference Candidates

Direction	Candidate	Source	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	`rcim_retuned_GBM19_Fw`	`rcim_retuned`	0.001089	0.001299	2.372	4.912
backward	`rcim_retuned_GBM19_Bw`	`rcim_retuned`	0.002766	0.003300	5.398	12.280

Repository-Owned Static Baselines

Direction	Current Strongest Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
forward	`tree_Fw`	0.003053	0.003395	6.731	11.995
backward	`tree_Bw`	0.003258	0.003651	7.051	14.116
global, forward side	`tree_global`	0.002998	0.003364	6.590	11.601
global, backward side	`tree_global`	0.003290	0.003702	7.118	13.703

The `tree` family is the current strongest repository-owned offline static baseline, but deployment suitability remains a separate decision because large tree artifacts can be PLC/TwinCAT-unfriendly.

Visual Verification Evidence

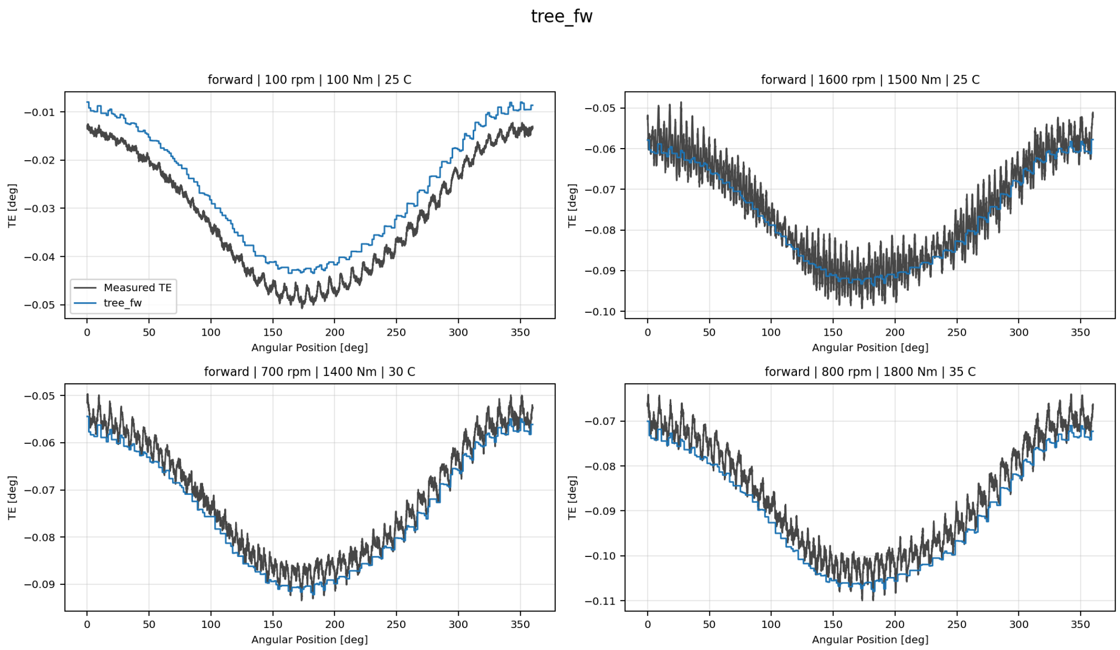
Best-Model Collage Evidence

The best-model collage report checks local tracking behavior candidate by candidate. Each collage contains four deterministic held-out test curves. The current refreshed bundle includes forward and backward reference collages, `Wave 1` family-best collages for `feedforward`, `harmonic_regression`, `periodic_mlp`, `residual_harmonic_mlp`, and `tree`, plus refreshed explicit-harmonic `periodic_mlp` campaign collages for `global`, `Fw`, and `Bw`.

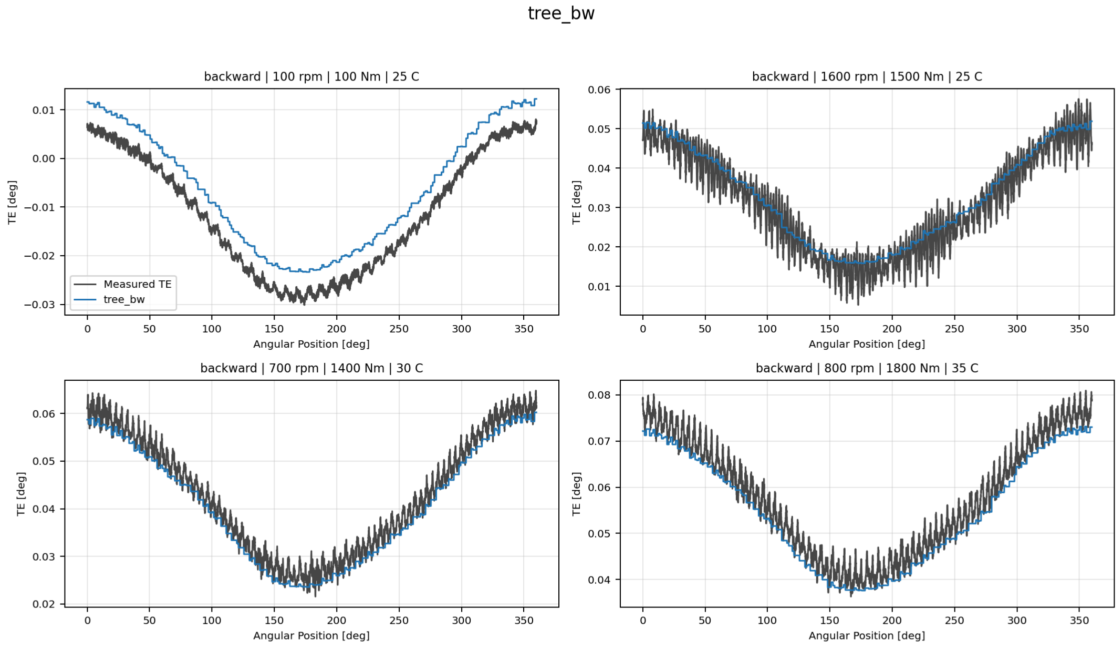
The PDF companion is the official visual appendix for candidate-level curve inspection:

`doc/reports/analysis/track2/best_model_collage_report/[2026-05-20]/track2_best_model_collage_report.pdf`

Representative evidence from the refreshed validation output:



Forward Wave 1 Tree Collage



Backward Wave 1 Tree Collage

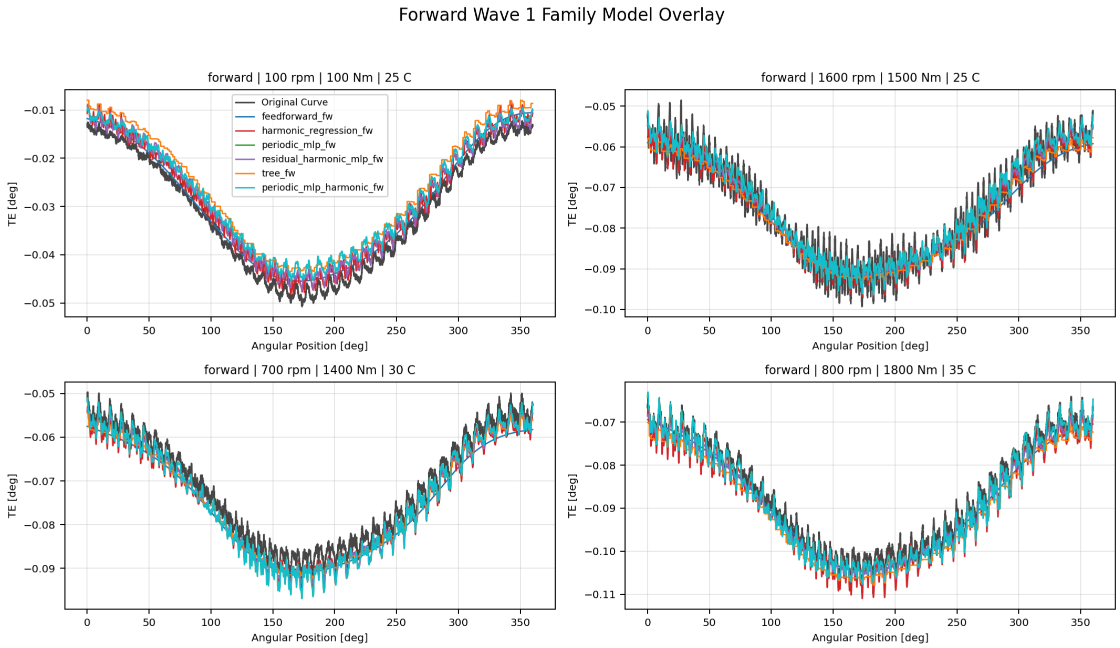
Multi-Model Overlay Evidence

The multi-model curve comparison report overlays several models on the same measured curve. This is the official visual appendix for relative curve tracking and family screening. The current refreshed bundle includes forward and backward reference overlays, forward and backward `Wave 1` family overlays, and screened `RCIM Model-Bank Reproduction` versus `Wave 1` overlays for both directions.

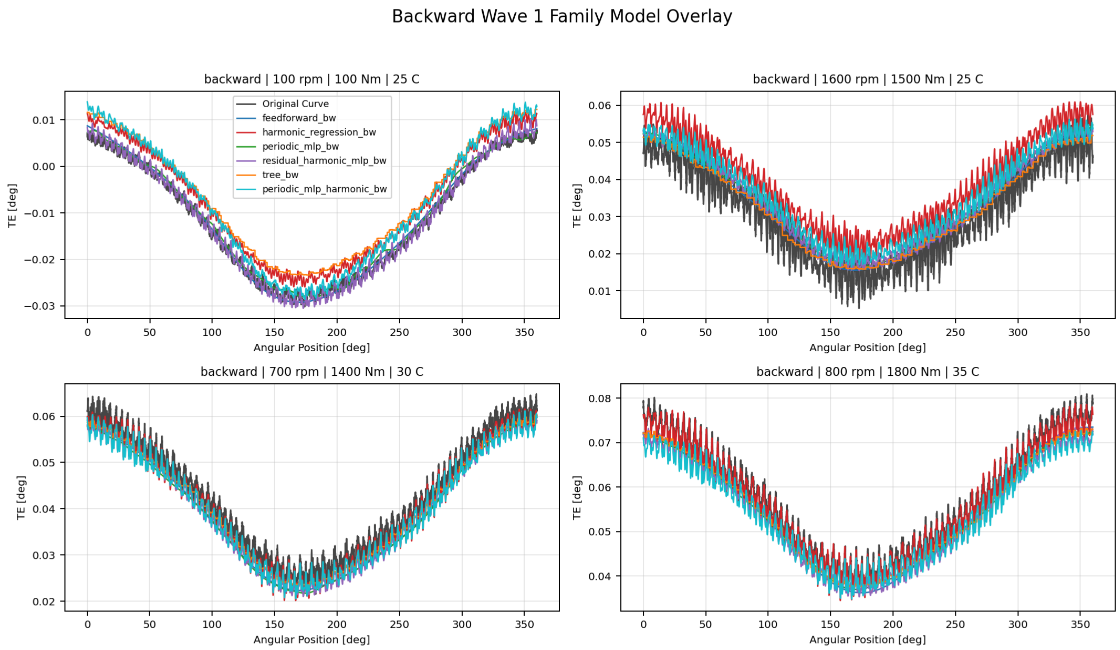
The PDF companion is

`doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-05-20]/track2_multi_model_curve_comparison_report.pdf`

Representative evidence from the refreshed validation output:



Forward Wave 1 Overlay



Backward Wave 1 Overlay

Campaign Update Ledger

Future `TE Curve Verification Pipeline` campaigns must append a row here after the campaign result report, matrix refresh, visual report refresh, and official PDF validation are complete.

Date	Campaign or Update	Candidate Scope	Matrix Status	Visual Status	Decision
2026-05-21	`periodic_mlp` explicit harmonic registry refresh	`global` , `Fw` , `Bw` visual verification	source matrix unchanged; visual registry refresh included	collage and overlay PDFs refreshed	included as visual evidence, not promoted over `tree`

Maintenance Contract

For every future `TE Curve Verification Pipeline` model-verification update:

1. refresh `Track 2 Directional Model Comparison.md` when the candidate set or numeric matrix changes;
2. regenerate the best-model collage report when a new family or promoted candidate needs local curve inspection;
3. regenerate the multi-model curve comparison report when relative visual screening changes;
4. append the campaign or update to this report ledger;
5. export and validate this official report PDF;
6. update `doc/running/te_model_live_backlog.md` ;
7. update `Training Results Master Summary.md` when the accepted current best status or campaign interpretation changes.

Closeout Decision

`TE Curve Verification Pipeline` is closed as the current official offline verification report. Its accepted baseline for future work is:

- forward paper-derived comparison: `paper_retuned_best_Fw` ;
- backward paper-derived comparison: `paper_retuned_best_Bw` ;
- strongest individual paper-reference family evidence: `rcim_retuned_GBM19_Fw` and `rcim_retuned_GBM19_Bw` ;
- strongest repository-owned static baseline: `tree` / `hist_gradient_boosting` . The next modeling branch can proceed to `Wave 2.1` temporal models. Online compensation remains outside `TE Curve Verification Pipeline` and is tracked under `Track 3` / `Target B` .